

Topology Assignment 1

Problem 1: Show that a subset is closed if and only if it contains all of its limit points

Solution: Define x to be a limit point of S if for all neighbourhoods $nb(x)$, we have $(nb(x) - \{x\}) \cap S \neq \emptyset$

Define the closure of S , as the smallest closed set that contains S . Denote it $cls_X(S)$

Define

$$L := \{x \in X : \forall nb(x), nb(x) \cap S \neq \emptyset\}$$

Lemma: L and $cls_X(S)$ are the same, i.e.,

$$\bigcap_{\alpha: S \subseteq F_\alpha} F_\alpha = \{x \in X : \forall nb(x), nb(x) \cap S \neq \emptyset\}$$

Proof. Let x not be in L , i.e., $\exists nb(x)$ such that $nb(x) \cap S = \emptyset$ or $nb(x) \subseteq S^c$ or $S \subseteq nb(x)^c$. This implies that $cls_X(S) \subseteq nb(x)^c$ which means that x is not in $cls_X(S)$.

Let x on the other hand, not be in $cls_X(S)$, which means $x \in (cls_X(S))^c$ which means that there exists a neighbourhood of x that does not intersect S , making it not in L . We are done. $\square$

Lemma: $cls_X(S) = S \cup S'$ where S' is the set of all limit points of S .

Proof. If x is a point of $cls_X(S)$, and is in S , then it trivially is in S . say $x \in cls_X(S)$ but isn't a point of S , then for every neighbourhood $nb(x)$, we have that $nb(x) - \{x\} \cap S \neq \emptyset$, making x a limit point of S . If x is a limit point of S , of course, it is obviously in $cls_X(S)$, and if it is in S , it obviously is in $cls_X(S)$. Hence, $cls_X(S) = S \cup S'$. $\square$

Lemma: A set is closed if and only if $A = cls_X(A)$

Proof. $\implies$) Let $A = cls_X(A)$. This means obviously that A is closed.

$\impliedby$) Suppose A is closed. Obviously, $A \subseteq \bigcup_{\alpha: A \subseteq F_\alpha} F_\alpha$, and A itself is a closed set that contains itself, hence $cls_X(A) = A$. $\square$

If A is closed, $A = A \cup A'$ which means that $A' \subseteq A$. If $A' \subseteq A$, then $A \cup A' \subseteq A \cup A = A$.

Problem 2: (a) Constant map is Continuous, (b) Composition of two continuous maps is continuous

Solution: (a) Consider $f : X \rightarrow Y$ be $f(x) = c$. Let C be any open set in Y . If it contains c , then $f^{-1}(C) = X$ which is open. If $c \notin C$, then $f^{-1}(C) = \emptyset$ which is also open.

(b) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two continuous functions. Consider $g \circ f : X \rightarrow Z$.

Let $G \in \mathcal{T}_Z$ be any open set in Z . $(g \circ f)^{-1}(G) = \{x \in X : g \circ f(x) \in G\} = f^{-1}(g^{-1}(G))$ which is ultimately open in $\mathcal{T}_X$. Hence it is continuous.

Problem 3: Let $\mathcal{T}_D$ be the discrete topology on $\mathbb{R}$ and $\mathcal{T}_E$ be the euclidean topology on $\mathbb{R}$. Does there exist an homeomorphism from $\mathcal{T}_D$ to $\mathcal{T}_E$?

Solution: Let $f : \mathbb{R}, \mathcal{T}_D \rightarrow \mathbb{R}, \mathcal{T}_E$ be a homeomorphism. Let $\{U_\alpha\}$ be a basis for $\mathcal{T}_E$, we conjecture that $\{V_\alpha = f^{-1}(U_\alpha)\}$ is a basis for $\mathcal{T}_D$. Let $x \in V_1 \cap V_2$. This is an open set in $\mathcal{T}_D$. $f(V_1 \cap V_2)$ is also an open set in $\mathcal{T}_E$ that contains $f(x)$ which means that there exists U_α such that $f(x) \in U_\alpha \subset f(U_1 \cup U_2)$. This implies $x \in f^{-1}(U_\alpha) \subseteq f^{-1}(U_1 \cup U_2)$. Moreover, let U be open in $\mathcal{T}_D$. $U = f^{-1}(f(U))$. Moreover, since f is continuous and bijective, with a continuous inverse, we have that $f(U)$ is also open in $\mathcal{T}_E$, or rather it is an arbitrary union of basis elements $\cup_\alpha U_\alpha$. This implies $f^{-1}(f(U))$ is an arbitrary union of elements of $\{f^{-1}(U_\alpha)\}$. Hence $\{V_\alpha = f^{-1}(U_\alpha)\}$ is a basis for $\mathcal{T}_D$ (True for any homeomorphism).

We know that $\mathcal{T}_E$ has countable basis, i.e, is 2nd countable. Therefore, the collection $\{f^{-1}(U_i)\}$ forms a basis for $\mathcal{T}_D$. But note that (fact from analysis) any infinite discrete metric space is not 2nd countable (since it is not separable / lindelof). To see this, consider all the singleton elements. This is an open cover of $\mathbb{R}$, but contains no countable subcover. Hence, such a homeomorphism cannot exist.

Problem 4: Cofinite topology and countable complement topology:

Let X be a set. Let U be open if and only if U^C is finite, $\emptyset$ or X . Of course $\emptyset$ and X are in the topology. Consider U_α be a collection of open sets. $\cup_\alpha U_\alpha$. its complement is $\cap_\alpha U_\alpha^C$. If their complements are all X , then the intersection is X . If their complements are all empty, then the intersection is empty. If their complements are all finite, then the intersection is also finite, topology is closed under arbitrary unions. Consider $U_1 \cap U_2 \cap \dots \cap U_k = \cap_{i=1}^k (U_i)$. Its complement is $\cup_{i=1}^k (U_i^C)$. If each complement is finite, its finite union is still finite. If even one of their complements is X , then the finite union is also X . Hence, closed under finite intersections. Hence it defines a topology.

If we replace finiteness with countable: X and $\emptyset$ are still in $\mathcal{T}$. Suppose that U_α are such that each of their complements are X or $\emptyset$ or countable. $(\cup_\alpha U_\alpha)^c = \cap_\alpha U_\alpha^c$ which is ultimately either $X, \emptyset$ or countable (atmost countable). Let $U_1, U_2 \dots U_l$ be a finite collection whose complements are atmost countable sets. Their intersection's complement is $\cup_{i=1}^l U_i^c$ which is a finite union of atmost countable sets, which is atmost countable. If even one of the complements is X , the complement of the intersection would be X as well. Hence, if we replace finite with atmost countable, we still have a topology.

Problem 5: Let $X = \mathbb{Z}$ and $S_{a,b} := a + b\mathbb{Z} = \{a + bz : z \in \mathbb{Z}\}$

(a) If $x \in \mathbb{Z}$, then $x \in S_{0,x}$ obviously. Let $x \in S_{a,b}$ and $S_{c,d}$. $x = a + qb = c + q'd$. $S_{a,b}$ and $S_{c,d}$ are said to meet at x . The arithmetic progression jumps every b elements for $S_{a,b}$, and jumps every d elements for $S_{c,d}$. Hence, $S_{x, lcm(b,d)}$ contains x and is contained in

both $S_{a,b}$ and $S_{c,d}$. Hence, the two conditions for bases are verified.

(b) The topology generated by $\mathcal{B}$ is all possible unions of elements of $\mathcal{B}$. Consider $S_{a,b}^C$. This can be written as the union: $\cup_{j=1}^{b-1} S_{a+j,b}$. Hence, every $S_{a,b}$ is clopen.

(c) Consider $\{-1, 1\}^c$. Let there be only finite primes, i.e, $p_1, p_2 \cdots p_q$. Note that, every element in $\mathbb{Z}$ that is not -1 or 1 , is a multiple of one of $p_1, p_2 \cdots p_q$. Therefore, every non unit $(-1, 1)$ element can be covered by S_{0,p_i} . Hence, $\{-1, 1\}^C = \cup_{i=1}^q (S_{0,p_i})$ Which implies $\{-1, 1\} = \cap_{i=1}^q (S_{0,p_i}^C)$. Since S_{0,p_i} are open, we conclude that $\{-1, 1\}$ is open. But every open set in this topology must necessarily be infinite or empty, since a union of elements of $S_{a,b}$ ought to be infinite. Hence, we conclude that there must be infinite primes.
